

IgG binder parent proteins

Combined evidence, conservation and reproducible analysis

Three search strategies | filtered exact and partial mappings

Measure	Result
Input candidate binders	974
Eligible binders	792
Main natural full-length parent sequences	624
Parents with an exact eligible binder	621
Exact / nonexact eligible binders	609 / 183
Cross-species exact conserved binders	146
Additional putative nonboundary B sequences	51

Interpretation: natural full length is supported by database provenance. Novel binding, affinity, evolutionary origin and independent genomic translation are not established by sequence mapping alone.

What the analysis establishes

The three sources contain 1,670 distinct whole-protein amino-acid sequences. After record, natural genomic provenance, sequence-quality and B-domain filters, 624 sequences form the main parent database. Of these, 621 contain at least one eligible candidate binder exactly. These are database-supported natural full-length sequence candidates, not experimentally verified wild-type proteins or demonstrated binders. Sequence identity can associate a binder with multiple possible parents; it cannot identify one historical source of origin.

Binder filtering

The input contains 974 candidate binder IDs. 36 contain nonstandard characters (all X in this dataset), 154 are shorter than 40 amino acids, and 8 fall in both groups. Exclusions therefore leave 792 eligible queries. Sequences were not repaired by deleting X or other characters. Six noncandidate reference entries in the source database remain archival and are not counted as candidate binders. Unique full-sequence hashes define proteins; all original accessions and source occurrences remain available.

Pfam-source annotation recovery

The Pfam-source UniProt accessions were retrieved directly, including secondary-accession resolution. 77 UniProt records were sequence-verified against the supplied parent sequences. EMBL protein cross-references were followed, and 79 matching cross-references were verified against the retrieved amino-acid sequence before transferring NCBI CDS, completeness or genomic annotations. Nonidentical cross-references and fragment flags were retained in a separate audit and were not used to rescue parent completeness.

Exact mapping and the importance of the parent filter

Among the eligible queries, 674 have an exact occurrence in the inclusive input database. In the filtered main database, 609 have an exact occurrence and 183 do not. 65 of the latter had an exact occurrence in an excluded parent. Main-set exact mapping comprises 20,168 distinct binder-parent pairs and 20,376 sites, including repeat copies. 598 queries overlap a supported B-domain alignment by at least 80% of query length. 11 exact-mapped queries lack that degree of overlap and are not counted in primary domain conservation.

Partial-match results

Minimum ungapped substitutions are categorized separately as 1, 2, 3, 4, and >4: {'1': 140, '>4': 15, '3': 3, '2': 17, '4': 8}. Identity is $100 \times (\text{query length} - \text{substitutions}) / \text{query length}$, at 100% query coverage. All same-length windows of every retained parent are compared, and every equally best site is preserved. 168 nonexact queries reach the descriptive near-match rule of $\geq 90\%$ identity and $\geq 80\%$ B-domain overlap. This rule identifies candidates for validation, not confirmed parental origin. 10 queries achieve a lower edit distance when gaps are allowed. Weak nearest neighbors remain in the audit even if they are not persuasive homologs.

Gap-aware comparison

Edlib HW alignment consumes the full query while leaving parent flanks free; a substitution, insertion, or deletion each costs one. All equally best parent sequences are retained. A traceback, CIGAR, gap columns, substitution count, aligned identity and returned optimal locations are provided for each tied parent. One traceback does not enumerate every equivalent alignment path. Gapped identity uses matches / alignment columns; it must not be compared to an ungapped percentage without considering the denominator. Edlib uses unit-cost edit distance, not a substitution matrix or BLAST significance model. Gapped results are a cross-check, not an IgG-binding prediction.

Domain search and additional homologs

A targeted search used 38 archived Pfam HMMs corresponding to models present in the prior CDD annotation, implemented with pyhmmmer 0.11.1. This is not a complete search of every Pfam family. PF02216.23 has 53 model positions and sequence/domain GA cutoffs of 27/27 bits; both must pass. Main B domains additionally cover $\geq 80\%$ of model positions and span ≥ 40 aligned amino acids. HMM alignment and envelope coordinates are both exported. The main parents contain 2,328 qualifying domain occurrences, representing 292 distinct predicted envelope sequences. 51 differ from every original listed query and are not simple containment-based boundary variants. These are additional putative B-domain-family sequences, not experimentally confirmed IgG binders. A non-containment difference can still reflect a boundary or alignment artifact; domain boundaries and activity require follow-up.

Species distribution and exact conservation

The main database represents 25 resolved microbial species; 622/624 parent sequences have a resolved species association. 146 binder IDs have an identical full-query sequence in supported B-domain sites across at least two species. 419 occur in at least two distinct whole-protein sequences, and 205 occur in at least two targeted profile contexts. Different whole-protein sequences can be alleles of the same biological protein, so these counts must not be presented as distinct protein families. Species counts are overlapping sets: a shared sequence contributes to each species where observed. These are properties of a search-enriched database, not unbiased species prevalence.

Exclusions applied in this revision

Species assignments to *Klebsiella pneumoniae* and *Streptomyces malaysiensis* were removed before parent provenance, species conservation and locus counts were recalculated. The eight additional non-Staphylococcaceae species listed in the exclusion policy were also excluded at occurrence level; their candidate sequences had already failed main-parent filters. Protein WMZ62091.1 was excluded because its B plus DNA_gyraseB_C context remains unresolved. Other *Staphylococcus pseudintermedius* parents were retained. Sequences independently supported by retained species were preserved, rather than deleting every sequence with one disputed occurrence. The original inputs and rejected occurrences remain archival evidence only. The revised primary and conservative cross-species counts both equal 146. A retained species label is database-supported, not an independently validated assembly taxonomy.

Genomic occurrences and natural-source evidence

NCBI Identical Protein Groups (IPG) expanded nonredundant WP accessions to genomic CDS occurrences. The main set has 109,451 resolved-species locus keys. Keys use nucleotide accession/version, coordinates and strand. An NZ_wrapper is normalized to its exact underlying INSDC accession. Other RefSeq/GenBank accession aliases are retained when no explicit equivalence is available, so locus counts can still overcount biological loci. Assembly versions and closely related strains are not independent biological replicates. IPG links are database annotation evidence: every genomic DNA sequence was not independently retranslated, and assemblies were not reassembled or checked for contamination. Do not label the locus total as an exact number of independent genes or organisms.

Completeness, engineered sequences and uncertainty

Full length requires an explicit record completeness claim or nonfuzzy CDS coding boundaries in a sequence-identical record. A Protein feature that simply spans the sequence is insufficient. Genome-wide pseudogene statistics in generic PGAP comments are ignored when assigning protein-specific sequence cautions. New sequence-identical records supersede completeness claims for the same accession in the old snapshot; discrepancies remain in the record table. Natural support requires microbial genomic occurrence or complete microbial CDS provenance. Synthetic constructs, engineered/fusion/mutant annotations and sequence cautions are reviewed at occurrence level. A matching independent natural occurrence can support a sequence also seen in a construct. Recombinant expression or synonymous codon optimization alone does not prove an amino-acid change. Natural plasmids are not automatically removed. Lack of an engineering keyword alone is not the natural-source criterion. Nonstandard residues, accession-sequence conflicts, uncertain completeness and lack of required domain support are explicit audit reasons. A retained record remains only as reliable as its source annotations.

Protein context and localization

The analysis combines recovered NCBI protein features, prior all-parent CDD results, and the archived targeted Pfam scan. Signal-peptide, mature-peptide and other available record features are exported with their qualifiers. YSIK signal, Gram_pos_anchor, LysM and Sbi-IV profile evidence is distinguished from an experimental localization result. LPXTG is reported as a literal motif with coordinates, not proof of sortase processing. SignalP and a dedicated transmembrane predictor were not run; absence of their predictions is not absence of a signal peptide or membrane segment. Novel architectures lacking support from these selected models may be missed. Perform a complete current InterPro/Pfam scan if the paper requires an exhaustive protein annotation claim.

Near-conservation and alignment

Exact cross-species sharing is kept separate from near-conserved clusters. Deterministic representative clustering uses global unit-cost alignments at 95% and 90% identity and at least 90% residue-pair coverage of both sequences. Membership is relative to the representative; not every pair of members necessarily meets the threshold. The tables report representative identity, coverage, members and observed species. Unique predicted B-domain envelope sequences are aligned to PF02216.23. Position occupancy, residue counts, consensus frequency and entropy are exported. The logo uses $(\log_2(20) - \text{residue entropy})$ multiplied by nongap occupancy. It has no small-sample correction and no phylogenetic weighting; sequence redundancy is reduced by exact deduplication, not eliminated evolutionarily.

Biological interpretation

Shared B-domain sequences in different whole-protein contexts are consistent with modular reuse, repeat duplication, or retention of related immunoglobulin-binding modules. This dataset alone cannot distinguish vertical inheritance, recent divergence, horizontal transfer, taxonomic error, or sampling effects. In experimentally studied *S. aureus*, Protein A and Sbi contain related immunoglobulin-binding domains, whereas Sbi also carries domains involved in complement interactions. Thus B plus Sbi-IV supports an Sbi-like context hypothesis, and B with cell-wall-associated profiles supports a different context hypothesis. These are annotation-based interpretations, not demonstrations that every newly identified sequence binds IgG, has the same affinity, or behaves similarly across host IgG species/subclasses. Even conserved residues may have different effects because of neighboring domains, accessibility, avidity and localization.

Figures appropriate for the paper

Use the exact overview as a dataset map; the complete matrix is intentionally dense and the indexed atlas supplies readable detail. Pair a species distribution plot with the species-by-conserved-binder heatmap. The species-sharing matrix gives an interpretable summary of identical domain sharing. The architecture chart separates sequence sharing across contexts from mere accession redundancy. Keep partial similarities in their separate overview and atlas with an explicit unconfirmed-parent note. A domain conservation logo is useful as supporting evidence, but it must not be described as a binding-affinity model. The full parent/site atlases, mapping matrices, row/column keys and complete tables belong in supplementary materials.

Direct response to the full-protein-context reviewer

The requested addition is a dataset-wide parent-context classification, rather than discussion of only two lead binders. The architecture classes and counts are provided in `parent_context_summary.tsv` and the workbook, with exact domain-supported parent counts distinguished from all retained search candidates. The classification table separates full-parent counts from binder counts, which overlap across classes. The representative-architecture figure is suitable for the main text; the complete table and indexed atlases belong in supplementary material. The comment itself does not demand additional binding experiments.

What is sufficient for a publication

For a computational parent-association and domain-conservation claim, these results provide traceable accessions, exact sequence hashes, full-query mappings, all ties, model thresholds, sensitivity analysis, source labels and exclusion audits. The manuscript must still supply the original search dates, database releases, program versions, query lists, parameters and TBLASTN nucleotide-to-protein extraction procedure; these cannot be reconstructed from the uploaded FASTA alone. If the central claim is a novel functional IgG binder, validate binding with appropriate positive/negative controls, concentration dependence and specificity, and determine affinity when relevant. If the claim is horizontal transfer or evolutionary origin, add phylogeny, genomic-neighborhood comparison, assembly quality review and taxonomic corroboration. If the claim is a natural full-length biological protein, targeted genomic/CDS validation and expression/localization evidence strengthen the database-supported conclusion.

Sensitivity and reproducibility

The sensitivity table compares domain HMM coverage thresholds of none, 60% and 80%, and binder overlap thresholds of 50%, 80% and 100%, holding the record/natural/sequence gate fixed. Qualifying B alignments must still span ≥ 40 aa. Archived XML, profile models, model versions, retrieval hashes, source-file hashes, coordinate conventions and all derived tables accompany the results. The original database is preserved inside the derived SQLite database. The source labels are NCBI protein BLAST, Pfam search, and TBLASTN search; no original search parameters are invented. Code is written for Python 3.9 and provided as a Jupyter workflow. The current execution used Python 3.12; a Python 3.9 interpreter was not available here, so syntax compatibility is checked separately and a compatible environment specification is supplied.

Draft Methods paragraph

Candidate sequences were filtered to at least 40 residues and the 20 standard amino acids. Parent sequences from NCBI protein BLAST, Pfam search and TBLASTN search were deduplicated by complete amino-acid sequence while retaining accession and source provenance. NCBI records were sequence-verified and expanded with IPG genomic occurrences and taxonomy. The primary set required evidence-supported natural provenance, complete-record/CDS annotation, standard residues, no accession-sequence conflict, and a PF02216.23 hit passing sequence and domain gathering cutoffs (27 and 27 bits), covering at least 80% of the 53-position model and at least 40 aligned residues. Species occurrences in the declared exclusion policy and the unresolved WMZ62091.1 parent sequence were excluded before downstream analysis. Independently supported occurrences of shared sequences in retained species were preserved. Every exact whole-query occurrence in retained parents was retained. Domain conservation required at least 80% query overlap with a supported B-domain alignment. Nonexact queries were compared against every same-length parent window, preserving all minimum-substitution ties, and cross-checked using full-query semi-global unit-cost edit alignment. Species sharing was calculated from identical complete query sequences in supported domain sites, separately from unique-parent-sequence and genomic-locus sharing. A sensitivity analysis varied domain coverage and overlap thresholds. Database-supported associations were not interpreted as experimental evidence of IgG binding.

References and source methods

Pfam gathering thresholds: <https://pfam-docs.readthedocs.io/en/latest/glossary.html>

PF02216 HMM retrieval: <https://www.ebi.ac.uk/interpro/api/entry/pfam/PF02216?annotation=hmm>

NCBI IPG data model: <https://www.ncbi.nlm.nih.gov/ipg/docs/about/>

NCBI nonredundant WP proteins: <https://www.ncbi.nlm.nih.gov/refseq/about/nonredundantproteins/>

PyHMMER methods: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10159651/>

Burman et al., Sbi immunoglobulin/complement domain experiments:
<https://pmc.ncbi.nlm.nih.gov/articles/PMC2649420/>

Smith et al., Sbi immune-evasion experiments: <https://pmc.ncbi.nlm.nih.gov/articles/PMC3165492/>